Aggregative and Contrastive Dual-View Graph Attention Network for Hyperspectral Image Classification

Haoyu Jing, Sensen Wu, Laifu Zhang, Fanen Meng, Tian Feng, Yiming Yan, Yuanyuan Wang, and Zhenhong Du, *Member, IEEE*

***Abstract*— Graph convolutional networks (GCNs) have recently gained prominence in hyperspectral images (HSIs) classification tasks given their superior performance on non-Euclidean data. However, GCN-based methods are heavily reliant on complete graph structural information, which can cause the aggregation and transmission of information across nodes from differing classes, thereby compromising the classification performance. Furthermore, the scarcity of labeled pixels in HSIs often limits the representational capability of such methods. To address these issues, we propose an aggregative and contrastive dual-view graph attention network (ACoD-GAT) for HSI classification. Specifically, we present a progressive aggregation module, including a pixel clustering submodule and a node aggregation submodule to exploit semantic information at various levels. Besides, we integrate multiscale manipulation with a diffusion matrix to construct the dual view to further extract semantic information from both local and global perspectives. Moreover, we design an unsupervised contrastive loss function and a supervised contrastive loss function to facilitate contrastive learning on the dual view, improving the representational capabilities of ACoD-GAT with very few labeled samples. The extensive experimental results on four benchmark datasets demonstrate the superiority of the proposed ACoD-GAT compared with other state-of-the-art methods.**

***Index Terms*— Contrastive learning, dual view, graph convolutional network (GCN), hyperspectral image (HSI) classification, progressive aggregation module (PAM).**

I. Introduction

HYPERSPECTRAL images (HSIs) comprise continuous and diverse spectral bands, providing abundant spatial and spectral information. HSIs have found widespread applications across various domains, including agricultural management [1], mineral identification [2], and environmental protection [3]. HSI classification has played an important role in these areas. However, due to factors such as atmospheric mixing effect and diversity of ground objects, a large amount of noise is introduced into HSIs, which poses a great challenge to the accurate classification of HSIs.

Over the last few decades, diverse methods have been proposed for HSI classification. The early staged methods mainly rely on conventional machine learning methods, such as k-nearest neighbor (KNN) [4], Bayesian methods [5], principal component analysis (PCA) [6], [7], random forest [8], extreme learning machine (ELM) [9], and kernel methods [10], which to some extent progress the development of HSI classification. Prominently, support vector machine (SVM) [11], [12], [13] has gained considerable attention due to their remarkable performance in datasets with scant labeled samples, but it comes with its inherent shortcomings, most notably its inability to harness spatial information within images. Addressing this limitation, Camps-Valls et al. [14] developed composite kernels adept at incorporating spatial information. Furthermore, the Markov random field (MRF) [15] has also demonstrated efficacy. MRF can utilize local spatial information and has strong spatial constraints. However, MRF is based on the simple smoothness and homogeneity assumption throughout the entire HSI, and its classification effect on irregular areas is limited. Moreover, these methods are quite empirical and require extensive handcrafted configurations.

In recent years, deep learning methods have garnered increasing attention in HSI classification due to their ability to automatically extract high-level representations [16], [17], [18], [19], [20], [21], such as deep belief networks (DBNs) [22], recurrent neural networks (RNNs) [23], [24], and generative adversarial networks (GANs) [25]. Convolutional neural networks (CNNs) have garnered significant attention given their characteristic of weight sharing, which requires fewer parameters. This trait makes CNNs particularly suitable for large-scale HSI applications. HSI classification methods based on CNN can be categorized into the following types based on the input information: spectral CNN, spatial CNN, and spectral–spatial CNN. The first category of CNNs processes each pixel spectral vector as input and uses a 1-D convolution kernel for feature extraction [26]. Spatial CNN, as proposed by Makantasis et al. [27], employing 2D-CNN

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Haoyu Jing, Sensen Wu, Laifu Zhang, Fanen Meng, Yiming Yan, and Zhenhong Du are with Zhejiang Provincial Key Laboratory of Geographic Information System (GIS), School of Earth Sciences, Zhejiang University, Hangzhou 310058, China (e-mail: 12138033@zju.edu.cn; wusensengis@zju.edu.cn; giserfu@zju.edu.cn; 12238036@zju.edu.cn; yanyiming@zju.edu.cn; duzhenhong@zju.edu.cn).

Tian Feng is with the School of Software Technology, Zhejiang University, Hangzhou 310027, China (e-mail: t.feng@zju.edu.cn).

Yuanyuan Wang is with Ocean Academy, Zhejiang University, Zhoushan 316021, China (e-mail: wangyuanyuanxy@zju.edu.cn).

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for spatial feature extraction. Spectral–spatial CNN aims to conduct feature extraction by leveraging both spectral and spatial features. For instance, Yang et al. [28] designed a 1-D + 2-D CNN to separately extract and then combine these features, while Li et al. [9] introduced a 3D-CNN that enhances model generalization through the integration of regularization technology. Zhong et al. [30] proposed a residual CNN to learn discriminative features with spectral and spatial residual blocks. While CNN-based methods have shown promising results, they encounter a limitation in capturing the sequence. When there are many categories to be distinguished and spectral features are similar, the performance of CNNs will encounter bottlenecks. Being aimed at this issue, Hong et al. [31] proposed a Transformer-based method called SpectralFormer, which can learn local detailed spectral representations and convey information from shallow layers to deep layers. In order to improve the ability to capture deep semantic features, Sun et al. [32] employed 2-D + 3-D convolution modules and transformer to extract shallow and deep features respectively. Xue et al. [33] proposed a local Transformer that dynamically obtains spatial attention weights by measuring the similarity between related pixel pairs.

However, existing deep learning approaches have demonstrated considerable success, and they necessitate extensive labeled samples. Yet, procuring such substantial samples in practical scenarios frequently poses challenges, and these deep learning techniques struggle to derive discriminative representations when confronted with limited labeled samples. Addressing this issue, numerous researchers have embarked on investigations, such as few-shot learning [34], [35] and contrastive learning [36], [37]. Contrastive learning obtains discriminative representations by measuring similarities and dissimilarities between unlabeled samples [38]. In HSI classification, there have been some works employing contrastive learning. Hou et al. [39] proposed a two-stage training strategy that employes contrastive learning to pretrain the model and then fine-tunes the model with a small number of labeled samples. Huang et al. [40] combined Swin Transformer [41] and contrastive learning to fully utilize the multiscale semantic information of HSI. Lu et al. [42] utilized a pairwise contrastive learning subnetwork to evaluate the distance between samples and fused multilevel spectral–spatial features to further improve the classification performance. Guan and Lam [43] introduced a progressive training framework to achieve strong initialization for pretraining.

Although the above methods have shown excellent performance, they cannot deliver end-to-end classification outcomes and ignore the topological information of the image, which hampers their ability to capture the geometric variations across different regions in HSIs. In contrast, graph convolutional networks (GCNs) [44], [45], [46] are suited for non-Euclidean data and operate based on predefined graphs, allowing for the passing, transforming, and aggregating of features across graph nodes. Moreover, as the GCN is a semi-supervised learning method, it can effectively aggregate information using both labeled and unlabeled samples simultaneously [47]. Given that GCNs can process non-Euclidean data with a graph structure, they are well suited for retaining class boundary information and have been applied in the HSI classification [48]. Mou et al. [49] constructed a graph structure with the original HSI, where pixels represent nodes and edges are determined by the node similarity. Liu et al. [50] merged irregular superpixel features encoded by GCN and regular region features encoded by CNN, allowing CNN and GCN to collaborate in a single network. Zhang et al. [51] proposed graph pooling operations to minimize redundant information in graphs. Ding et al. [52] proposed autoregressive moving average filters to capture global graph structures. Tong et al. [53] defined the feature aggregation function by introducing an attention weighted graph. Yu et al. [54] proposed a GCN based on edge inference, which simultaneously exploits instance- and distribution-level similarities to infer relationships between samples. Wan et al. [55] developed a dynamic GCN that refines the adjacency matrix progressively. Hong et al. [56] mitigated the computational demands linked with the adjacency matrix by randomly selecting nodes to constitute a subgraph and employed mini-batch training to expedite the process of gradient descent. Xue et al. [57] employed the differentiated-scale graph construction method to better model the spatial structure of HSI and integrated a Siamese network with GCN to remedy the pixel-level feature. In conventional GCNs, the weights between nodes are static, limiting the network's expressive power. To overcome this, the graph attention networks (GATs) [58] were introduced, allowing dynamic adjustment of weights between nodes, enhancing its application in HSI classification [59]. Zhao et al. [60] took all samples as nodes to construct the graph and employed GAT to perform semi-supervised classification. Pu et al. [61] combined spectral filtering and GAT to reduce the redundancy of spectral feature. Li et al. [62] proposed a multistage graph construction strategy to fully utilize the spatial topology and employed the sparse GAT to trim the graph. However, existing methods still have some limitations.

1) In current studies, the construction of graph structures concentrates on single-tier semantics, namely, pixel or object level, neglecting a holistic consideration incorporating pixel-, object-, and region-level semantics simultaneously.
2) Besides, existing methods require the input of complete graph structure information during training, which leads to the information aggregation and transmission across a large number of nodes from different classes, resulting in impaired discrimination ability and lower classification accuracy.
3) Due to their semi-supervised paradigm, existing GCN-based methods can demonstrate decent performance with relatively fewer labels. However, their effectiveness remains restricted in instances with severely limited samples.

To tackle the problems above, this article proposes an aggregative and contrastive dual-view GAT (ACoD-GAT) for HSI classification. In order to establish connections across different levels of HSI semantics, we design a progressive aggregation module (PAM), which obtains regions containing similar nodes. We construct the dual view and employ GAT to acquire the node representation, which can prevent the

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aggregation and transfer of information from different class of nodes as much as possible. In addition, to fully utilize limited labeled samples and exploit the abundant information in numerous unlabeled samples, we incorporate contrastive learning into our model. We design both a supervised contrastive loss (SCL) function and an unsupervised contrastive loss (UCL) function to significantly enhance the model's representational capability. To sum up, the main contributions of this article are given as follows.

1) A PAM is proposed to progressively establish semantic relation between pixels, objects, and regions, utilizing various levels of HSI contextual information. PAM comprises two important modules: the pixel clustering submodule (PCM), which aggregates pixels into superpixels, and the node aggregation submodule (NAM), which treated superpixels as nodes to construct graph and group these nodes to form regions predominantly composed of nodes from the same class.
2) We construct the dual view for each region that includes both a local view and a global view to fully capture contextual information at different scales and employ GAT to obtain the node represents of dual view of each region. This design effectively improves the second point of the aforementioned limitation of the existing method.
3) We design both a supervised contrastive loss (SCL) function and an UCL function to maximize the mutual information between identical nodes or nodes of the same class across the dual view. This approach leverages self-supervised information and limited sample data to enhance intraclass compactness and interclass separability.

The reminder of this article is organized as follows. Section II introduces the proposed ACoD-GAT in detail. Section III validates and analyzes the proposed method on four public datasets. Section IV concludes this article.

II. Proposed Method

In this section, the proposed ACoD-GAT method is described. First, we provide a brief explanation of the architecture. Then, we carefully describe the PAM, explaining the specific implementation details of PCM and NAM. Next, we formulate a dual-view construction method and a regional training strategy. Finally, we describe the designed contrastive loss function from two aspects: unsupervised and supervised.

The pipeline of ACoD-GAT is shown in Fig. 1. The PAM receives HSI and extracts its useful semantics at different levels. Specifically, the PCM clusters pixels in HSI into superpixels. The NAM constructs a graph by employing superpixel nodes and aggregates nodes into multiple regions (see Section II-A). The ACoD-GAT extracts the subgraphs corresponding to each region and then obtains the dual view through multiscale manipulation and calculating the diffusion matrix. After obtaining the dual view, GAT is used to obtain the corresponding node representation (see Section II-B). Finally, ACoD-GAT utilizes contrastive learning to enhance representation and discrimination abilities (see Section II-C).

A. Progressive Aggregation Module

HSIs contain rich multilevel semantic information, but most existing GCN models capitalize on semantic information at only a single level, which may impose performance constraints. To address this limitation, we introduce the PAM, comprising PCM and NAM, designed to progressively extract pixel-level, object-level, and region-level semantic information from HSI.

1) Pixel Cluster Submodule: The architecture of PCM is shown in Fig. 2. PCM comprises a series of feature extraction modules (FEMs) and a convolutional layer serving as the output layer. The FEM is employed to extract pixel-level features, incorporating elements such as reflection padding, 2-D convolution, instance normalization, and an activation function. Our objective is to cluster all pixels into superpixels $S = \{s_1, s_2, \ldots, s_n\}$, where s_n denotes the nth superpixel.

We conceptualize pixel clustering as an n-class classification problem. The initial input is an HSI denoted as $I \in R^{H\times W\times C}$ with height H, width W, and the number of bands C. However, the convolution operation is inherently translation-invariant and solely allocates superpixels according to localized spatial patterns. If only an HSI is given as input, independent connected components will be grouped into the same superpixel. Therefore, pixel locations $(x, y) \in R^{H\times W\times 2}$ and I are concatenated as the input. The output of output layer is denoted by $O_{\text{PCM}} \in R^{H\times W\times(C+N)}$. Reconstructed HSI $\hat{I} \in R^{H\times W\times C}$ and probabilistic representation of superpixels $P \in R^{H\times W\times N}$ can be obtained by splitting O_{PCM}. For all pixel positions (h, w), $\sum_n P_{h,w,n} = 1$, and the superpixel resulting from pixel clustering at position (h, w) is determined by $\text{argmax}_n P_{h,w,n}$.

Pixel clustering is inherently an unsupervised process. To facilitate this, we devise an unsupervised clustering loss L_{cluster} predicated on information entropy principles. This loss function aims to minimize the uncertainty inherent in pixel allocation. L_{cluster} is defined as follows:

$$L_{\text{cluster}} = -\frac{1}{\text{HW}}\sum_{n}\sum_{h,w} P_{h,w,n}\log P_{h,w,n}. \tag{1}$$

According to the clustering assumption [63], the decision boundary for the clustering model should pass through regions of low data density to prevent the bifurcation of data points within dense clusters. Due to the fact that information entropy can be reduced by simply reducing the number of classes, reliance solely on L_{cluster} may result in suboptimal clustering outcomes.

To mitigate the problem above, we design a balance loss L_{balance}, which can be written as

$$L_{\text{balance}} = \sum_{n} \hat{P}_n \log \hat{P}_n \tag{2}$$

where $\hat{P}_n = (1/HW)\sum_{h,w} P_{h,w,n}$ and $\sum_n \hat{P}_n = 1$. $\hat{P}_n$ represents the average probability that all pixels belong to s_n. L_{balance} is also defined based on the information entropy principles. When $\hat{P}_n$ obeys the uniform distribution, L_{balance} will approach its minimum value. This implies that minimizing L_{balance} can encourage PCM to generate uniformly sized superpixels.

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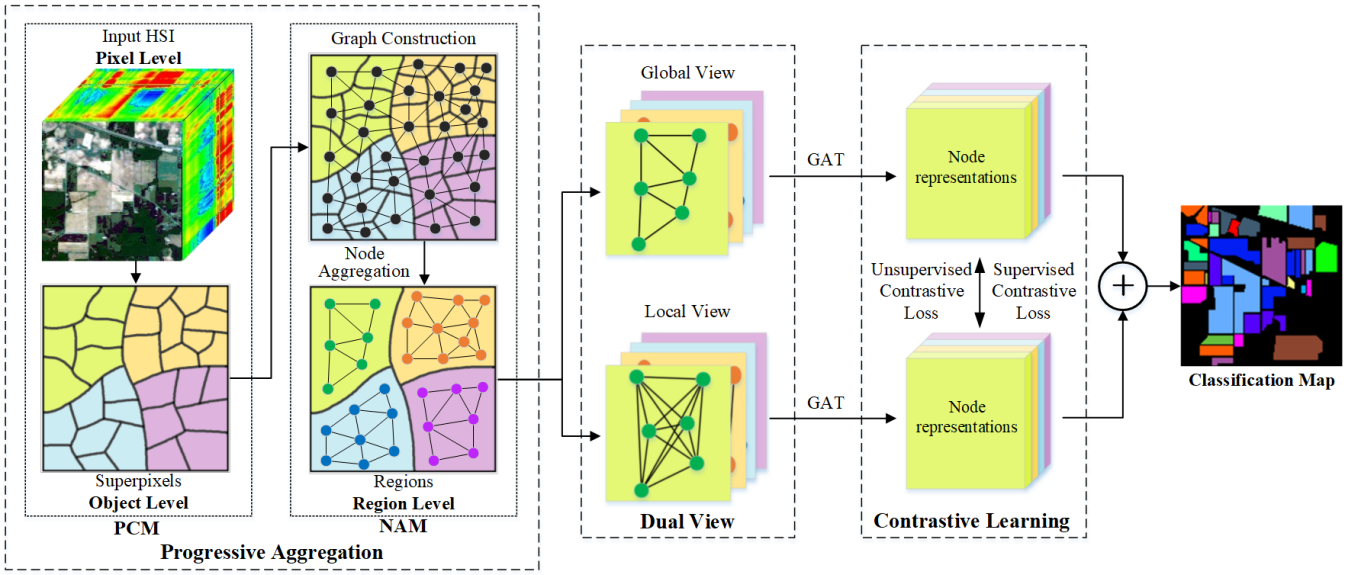

Fig. 1. Pipeline of our proposed ACoD-GAT. In the PAM, HSI is input into the PCM to generate superpixels containing object-level semantic information, where different colors represent different land-cover classes. The NAM constructs the graph according to the adjacency relation of superpixel nodes and aggregates nodes into different regions according to the topological properties and node attributes. In the dual-view construction phase, the local view and the global view are obtained by multiscale manipulation and calculating diffusion matrix on subgraphs corresponding to regions. The ACoD-GAT performs contrastive learning on the node representations of dual view obtained by GAT. The classification result is obtained by integrating the node representations of the dual view.

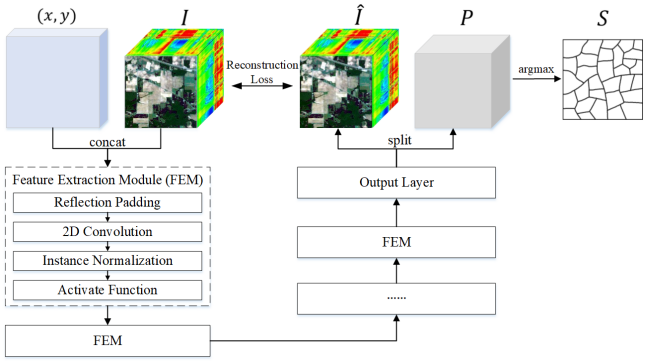

Fig. 2. Architecture of PCM.

In order to control the smoothness of superpixels, we posit a smoothness loss to impose penalties on gradients. We employ an edge-aware weighting derived from gradient magnitudes to define the smoothness loss $L_{\text{smoothness}}$ as follows:

$$L_{\text{smoothness}} = \frac{1}{\text{HW}} \sum_{h,w} \Big(\left\| \nabla_x P_{h,w} \right\|_1 e^{-\left\| \nabla_x I_{h,w} \right\|_2^2} + \left\| \nabla_y P_{h,w} \right\|_1 e^{-\left\| \nabla_y I_{h,w} \right\|_2^2} \Big) \quad (3)$$

where ∇ and $\|\cdot\|_p$ denote the gradients and p-norm, respectively, and $P_{h,w}$ and $I_{h,w}$ are the values of P and I at (h, w), respectively.

Finally, in order to preserve the structural information during the training process and fit superpixels to the detail components of the image, we consider that the module outputs both the P and the reconstructed HSI $\hat{I} \in R^{H \times W \times C}$ and define the reconstruction loss L_{recon} as follows:

$$L_{\text{recon}} = \frac{1}{\text{HWC}} \sum_{h,w} \left\| I_{h,w} - \hat{I}_{h,w} \right\|_2^2. \quad (4)$$

The overall loss of the PCM L_{PCM} is as follows:

$$L_{\text{PCM}} = L_{\text{cluster}} + \lambda_{\text{balance}} L_{\text{balance}} + \lambda_{\text{smoothness}} L_{\text{smoothness}} + \lambda_{\text{recon}} L_{\text{recon}} \quad (5)$$

where $\lambda_i, i \in$ balance, smoothness, recon is used to control the importance of each term. The characteristics of the produced superpixels can be adjusted by modifying λ_i.

The process of PCM is shown in Algorithm 1.

Algorithm 1 Pixel Clustering Process

1: **Input:** An HSI $I \in R^{H \times W \times C}$; pixel locations (x, y); hyperparameters: number of iterations T.
2: **for** $t = 1, \ldots, T$ **do**
3: $\left(P, \hat{I}\right) \leftarrow$ Get probability and reconstructed HSI by PCM with I and (x, y).
4: Calculate L_{PCM}.
5: Update parameters by gradient descent.
6: **end for**
7: **Output:**Get superpixels through $argmax_n P_{h,w,n}$.

2) Node Aggregation Submodule: In GCNs, fostering connections between nodes with analogous attributes can enhance the aggregation of features, as it introduces more pertinent information [64]. Conversely, interlinking nodes from disparate classes can obfuscate the distinction in node embeddings, thereby complicating classification tasks [65], [66], [67].

To address this issue, we introduce the NAM, which aggregates nodes into different regions according to their attribute characteristics and topological connection relation, often resulting in nodes within the same region sharing one class. Specifically, after obtaining the superpixels from PCM, we treat each superpixel as a node, with the mean spectral

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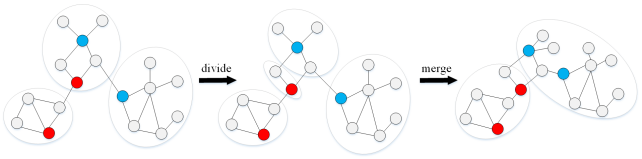

Fig. 3. Illustration of further processing of the preliminary results of node aggregation. Colored nodes represent labeled nodes, while different colors represent different classes.

features of the constituent pixels serving as the node's feature vector. In our proposed method, an undirected graph is denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ represents the set of nodes and $\mathcal{E}$ represents the set of edges. Consequently, we define $\mathcal{V} = \{v_1, \ldots, v_n,\ v_i \in R^C\}$, and employ cosine similarity $\text{sim}(a, b) = ((a \cdot b)/ \parallel a \parallel_2 \parallel b \parallel_2)$ to calculate the adjacency matrix A as follows:

$$A_{ij} = \begin{cases} \text{sim}(v_i, v_j), & \text{if } v_i \in \mathcal{N}_1(v_j) \text{ or } v_j \in \mathcal{N}_1(v_i) \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where $\mathcal{N}_1(v_i)$ is the one-order neighbor set of v_i. The merit of constructing an adjacency matrix in this way is that it can simultaneously integrate structural and node attribute information, which is beneficial for subsequent node aggregation work.

NAM aims to aggregate nodes in V into N_R regions $\boldsymbol{Reg} = \{\text{Reg}_i\}_{i=1}^{N_R}$. It is worth noting that N_R and the size of each region will be automatically determined during the NAM process without the need for manual settings. We refer to the work of network analysis [68] and define that $\boldsymbol{Reg}$ should have the following two properties under ideal conditions.

1) *Structural Closeness:* Nodes within a region exhibit structural proximity, whereas those in separate regions do not.
2) *Attribute Homogeneity:* Nodes within a region have homogeneous attributes, while nodes in different regions do not.

Therefore, we employ the Leiden algorithm [69] to aggregate nodes into regions and introduced modularity M to guide the aggregation process. The formula of M is given as follows:

$$M = \frac{1}{\sum_{i,j} A_{ij}} \sum_{i,j} \left[A_{ij} - \frac{\sum_i A_{ij} \sum_j A_{ij}}{\sum_{i,j} A_{ij}} \right] \delta(\text{Reg}_i, \text{Reg}_j) \tag{7}$$

where $m = \sum_{i,j} A_{ij}$, $k_i = \sum_j A_{ij}$, and Reg_i represents the region to which v_i belongs. When $\text{Reg}_i = \text{Reg}_j$, $\delta(\text{Reg}_i, \text{Reg}_j)$ is 1; otherwise, it is 0. By running the Leiden algorithm on G, we can obtain preliminary regions $\boldsymbol{Reg}'$ that maximizes M. As the construction of A relies on node attribute similarity, maximizing M will not only encourage structural similarity but also attribute homogeneity.

However, it should be noted that the Leiden algorithm only considers the topological information and fails to incorporate the category information of the nodes. To address this issue, we further improve the algorithm to facilitate semi-supervised node aggregation. Specifically, we have integrated two stages to refine $\boldsymbol{Reg}'$, as depicted in Fig. 3.

The first stage is region division. The purpose of this step is to further segment the regions in $\boldsymbol{Reg}'$ that contain nodes from different categories. Specifically, initialize $\boldsymbol{Reg}$ as an empty queue. Traverse $\boldsymbol{Reg}'$, and for each region r, perform the following assessments. If r does not contain labeled nodes of different classes, incorporate r into $\boldsymbol{Reg}$. If r contains labeled nodes from different classes, consider each labeled node and its adjacent nodes as a new region, add them to a queue Q, and mark the remaining nodes as unassigned. Then, repeat the following steps until Q is empty.

1) Get a node v_i from Q, find the unlabeled and most similar node v_j from adjacent nodes, and assign v_j to Reg_i.
2) Add unassigned and adjacent nodes of v_j to Q.

After the region division, regions will no longer contain labeled nodes of different classes, and newly generated regions are added to $\boldsymbol{Reg}$. The second stage is region merging. The purpose of this stage is to merge adjacent regions with nodes of the same class. The following steps are repeated until there are no alterations in $\boldsymbol{Reg}$.

1) Get a region from $\boldsymbol{Reg}$ and find adjacent regions. If these regions contain labeled nodes of the same class, merge these regions and remove them from $\boldsymbol{Reg}$.
2) Add the newly merged region to $\boldsymbol{Reg}$.

The detailed NAM optimization process is shown in Algorithm 2.

B. Dual-View Construction

In recent years, multiview learning has proven its significant role in learning superior representations [70]. The core concept of multiview learning is that an object or phenomenon can be depicted from multiple perspectives, such as varying feature types or different contexts. Each view provides insights into a specific aspect of the object. By integrating information from multiple views, a more comprehensive and rich expression of information can be obtained, thus improving model performance. However, research by Hassani and Khasahmadi [71] demonstrates that in graph learning, exceeding two views fails to enhance model performance while significantly augmenting computational load. For HSI, spatial contextual information is crucial. Most existing GCN models capture only local correlations among adjacent nodes, overlooking broader contextual information. To address this issue, we construct a dual view with local view and a global view of each region to fully utilize the spatial contextual information and generate a more comprehensive representation.

1) Local View: Constructing the local view by integrating semantic information across various scales is beneficial for fully utilizing the rich local information of HSI. Specifically, at scale s, each node x_i connects to its s-order neighbors, as illustrated in Fig. 4. The s-order neighbors of v_i can be expressed as follows:

$$\mathcal{N}_s(v_i) = \mathcal{N}_{s-1}(v_i) \cup \mathcal{N}_1(\mathcal{N}_{s-1}(v_i)) \tag{8}$$

where $\mathcal{N}_s(v_i)$ is the s-order neighbors set of v_i and $\mathcal{N}_0(v_i) = v_i$. In this article, we use two-order neighbors to construct the local view and use GAT to obtain node representations for

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Algorithm 2 Node Aggregation Process

Input: Node set V; labeled node set V'; adjacency matrix A.
Obtain $\boldsymbol{Reg}'$ calculated through the Leiden algorithm.
Initialize $\boldsymbol{Reg}$ as an empty queue.
for region r in $\boldsymbol{Reg}'$ **do:**
 if $size(unique(labelof(V' \cap r))) < 2$ **then**:
  Put r into $\boldsymbol{Reg}$.
 else:
  Initialize the temporary region set $\boldsymbol{tReg}$ as an empty queue.
  for node v in $V' \cap r$ **do**:
   Put v and nodes in $N_1(v) \cap r$ into a new queue Q.
   Put a new region consists of nodes above into $\boldsymbol{tReg}$.
  end for
  Put nodes in $r - Q$ into a list l.
  while Q is not empty **do**:
   Get a node v_i from Q and remove it from Q.
   Get the node $v_j = argmax(A_{ij}) \cap l$.
   Assign v_j to Reg_iin$\boldsymbol{tReg}$ and remove v_j from l.
   Put nodes in $N_1(v_j) \cap l$ into Q.
  end while
  Put regions in $\boldsymbol{tReg}$ into $\boldsymbol{Reg}$.
 end if
end for
while True **do**:
Flag $= 0$.
 for region r_i in $\boldsymbol{Reg}$ **do**:
  for region r_j in $(\boldsymbol{Reg} - r_i)$ **do**:
   if r_i is connected to r_j **and** the class of labeled nodes in r_i is same as in r_j **then**:
    Merge r_i and r_j as a new region r', flag $= 1$.
    Remove r_i and r_j from $\boldsymbol{Reg}$, put r' into $\boldsymbol{Reg}$, break;
   end if
  end for
  if flag $= 1$ **then**: break; **end if**
 end for
  if flag $= 0$ **then**: break; **end if**
end while
Output: Regions $\boldsymbol{Reg}$.

each region as follows:

$$Z^r = \sum_{s}^{2} \mathrm{GAT}_s\left(\mathcal{V}^r, A_s^r\right) \tag{9}$$

where GAT_s is the GAT encoder corresponding to the sth order neighborhood, $\mathcal{V}^r$ is the node set belonging to the rth region Reg_r, and A_s^r is the adjacency matrix of Reg_r at the sth order.

2) Global View: Considering that regions distant from each other in HSI may share the same landcover class despite not being directly connected, relying solely on neighborhood topology for information extraction is insufficient. Graph diffusion [72] is beneficial for transcending the limitations of the original topology by integrating contextual information from various neighborhoods. We utilize Personalized PageRank (PPR) [73] to construct the global view as follows:

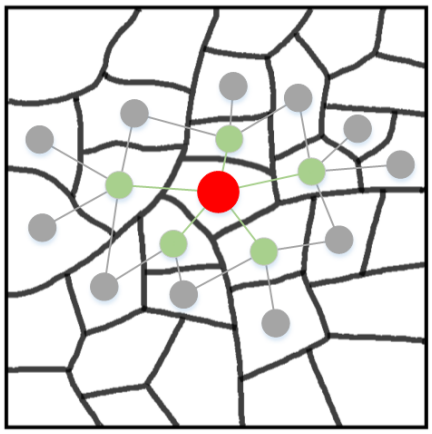

Fig. 4. Illustration of multiscale connections, using two-order neighbors as an example. The central node, one-order nodes, and two-order nodes are represented in red, green, and gray, respectively.

$$U = \beta\left(I - (1-\beta)D^{-\frac{1}{2}}\mathrm{AD}^{-\frac{1}{2}}\right)^{-1} \tag{10}$$

where $\beta \in (0, 1)$, I denotes the identity matrix, and D denotes the degree matrix.

The node representation of the global view obtained using GAT on Reg_r is as follows:

$$\tilde{Z}^r = \mathrm{GAT}_{\mathrm{global}}(\mathcal{V}^r, U^r) \tag{11}$$

where $\mathrm{GAT}_{\mathrm{global}}$ is the GAT encoder corresponding to the global view and U^r belongs to the U subset of Reg_r.

C. Loss Function

1) UCL Function: The dual view encapsulates both local information and global information of the HSI. To maximize the consistency between representations of the same node derived from both the local and global views, we design a UCL function L_{uc}. The formula for calculating the unsupervised loss L_{uc}^r on Reg_r is as follows:

$$L_{\mathrm{uc}}^r = \frac{1}{N_r}\sum_{i=1}^{N_r}\left(\frac{\left(\sum_{j=1}^{N_r}\exp\left(\mathrm{sim}\left(Z_i, \tilde{Z}_j\right)/\tau\right)\right)^q}{q} - \frac{\left(\exp\left(\mathrm{sim}\left(Z_i, \tilde{Z}_i\right)/\tau\right)\right)^q}{q}\right) \tag{12}$$

where $q \in (0, 1]$, N_r represents the number of nodes in Reg_r, and τ is the temperature coefficient.

2) SCL Function: To fully harness valuable labeled information, we develop an SCL function L_{sc}, aimed at maximizing the consistency of node representations within the same class. The formula for calculating the supervised loss L_{sc}^r on Reg_r

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is as follows:

$$L_{\mathrm{sc}}^{r}=\frac{1}{N_{r,l}}\sum_{i=1}^{N_{r,l}}\left(\frac{\left(\sum_{j=1}^{N_r}\exp\left(\mathrm{sim}\left(Z_i,\tilde{Z}_j\right)/\tau\right)\right)^{q}}{q}-\frac{\left(\sum_{k=1}^{N_{r,l}}\mathbf{1}_{[y_i=y_k]}\exp\left(\mathrm{sim}\left(Z_i,\tilde{Z}_k\right)/\tau\right)\right)^{q}}{q}\right) \tag{13}$$

where $N_{r,l}$ represents the number of nodes labeled on Reg_r, if the argument inside the bracket holds $\mathbf{1}_{[\cdot]}$ equals to 1, else 0. The class of x_i is represented as y_i, nodes belonging to the same class are considered positive node pairs, while nodes of different classes are considered negative node pairs.

3) Cross-Entropy Loss Function: After obtaining the graph representations of local view and global view, the final output on Reg_r can be represented as $\boldsymbol{Z}=\lambda_v Z^r+(1-\lambda_v)\tilde{Z}^r$, where $\lambda_v\in(0,1)$ denotes the weight assigned to Z^r. The cross-entropy loss function is used to penalize the difference between $\boldsymbol{Z}$ and labels, and the formula is given as follows:

$$L_{\mathrm{ce}}^{r}=-\sum_{i=1}^{N_{r,l}}\sum_{j=1}^{N_c}Y_{ij}\log\boldsymbol{Z}_{ij} \tag{14}$$

where Y denotes the label and N_c denotes the number of classes.

The overall loss $\boldsymbol{L}$ calculated on Reg_r can be expressed as

$$\boldsymbol{L}=\lambda_{\mathrm{uc}}L_{\mathrm{uc}}^{r}+\lambda_{\mathrm{sc}}L_{\mathrm{sc}}^{r}+\lambda_{\mathrm{ce}}L_{\mathrm{ce}}^{r} \tag{15}$$

where are hyperparameters used to adjust the weights of each loss function separately.

The process of ACoD-GAT is shown in Algorithm 3.

Algorithm 3 Process of the Proposed ACoD-GAT

1: **Input:** An HSI $I\in R^{H\times W\times C}$; label Y, number of iterations $\boldsymbol{T}$.
2: ***Reg*** $\leftarrow$ Get N_R regions by input I into PAM.
3: **for** $i=1,\ldots,N_R$ **do**
4: Generate local view and global view for region r_i.
5: **end for**
6: **for** $t=1,\ldots,\boldsymbol{T}$ **do**
7: **for** $r=1,\ldots,N_R$ **do**
8: Obtain Z^r and $\tilde{Z}^r$ via formula (9) and (12) respectively.
9: Input Y and calculate the overall loss $\boldsymbol{L}$.
10: Update network parameters via operating back propagation.
11: **end for**
12: **end for**
13: **Output:** Predict labels based on the trained network.

III. Experimental Results

A. Datasets

The performance of our proposed ACoD-GAT is evaluated on four real-world benchmark datasets, i.e., the University of Pavia, the Indian Pines, the Kennedy Space Center, and the Salinas, which will be introduced in the following.

TABLE I
Number of Training and Testing Samples for the University of Pavia Dataset

ID	Name	Train	Test
1	Asphalt	10	6621
2	Meadows	10	18639
3	Gravel	10	2089
4	Trees	10	3054
5	Painted metal sheets	10	1335
6	Bare soil	10	5019
7	Bitumen	10	1320
8	Self-blocking bricks	10	3672
9	Shadows	10	937

TABLE II
Number of Training and Testing Samples for Indian Pines Dataset

ID	Name	Train	Test
1	Alfalfa	10	36
2	Corn-notill	10	1418
3	Corn-mintill	10	820
4	Corn	10	227
5	Grass-pasture	10	473
6	Grass-trees	10	720
7	Grass-pasture-mowed	10	18
8	Hay-windrowed	10	468
9	Oats	10	10
10	Soybean-notill	10	962
11	Soybean-mintill	10	2445
12	Soybean-clean	10	583
13	Wheat	10	195
14	Woods	10	1255
15	Buildings-grass-trees-drives	10	376
16	Stone-steel-towers	10	83

1) University of Pavia: The University of Pavia dataset was captured in 2001 using the ROSIS sensor at the University of Pavia, Italy. Consisting of 610 × 340 pixels, this dataset displays a spatial resolution of 1.3 × 1.3 m and includes 103 spectral channels that encompass a wavelength spectrum between 0.43 and 0.86 μm. This dataset includes nine land-cover classes. Table I shows the samples used for training and testing for each class.

2) Indian Pines: The Indian Pines dataset was acquired by Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in northwestern Indiana, USA. Consisting of 145 × 145 pixels, the dataset, with a spatial resolution of 20 × 20 m, comprises 220 spectral channels. These channels span a continuous wavelength spectrum from 0.4 to 2.5 μm. Before any experiments are conducted, it is required to remove 20 water absorption and noise bands (104–108, 150–163, 220). The ground-truth map of this dataset includes 16 land-cover classes. Table II shows the pixels used for training and testing for each class.

3) Salinas: This dataset was acquired by the AVIRIS sensor over the Salinas Valley, USA. The dataset consists of 512 × 217 pixels with a spatial resolution of 3.7 m/pixel. After removing 20 water absorption bands, there remained a total

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TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE SALINAS DATASET

ID	Name	Train	Test
1	Brocoli_green_weeds_1	10	1999
2	Brocoli_green_weeds_2	10	3716
3	Fallow	10	1966
4	Fallow_rough_plow	10	1384
5	Fallow_smooth	10	2668
6	Stubble	10	3949
7	Celery	10	3569
8	Grapes_untrained	10	11261
9	Soil_vinyard_develop	10	6193
10	Corn_senesced_green_weeds	10	3268
11	Lettuce_romaine_4wk	10	1058
12	Lettuce_romaine_5wk	10	1917
13	Lettuce_romaine_6wk	10	906
14	Lettuce_romaine_7wk	10	1060
15	Vinyard_untrained	10	7258
16	Vinyard_vertical_trellis	10	1797

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE KENNEDY SPACE CENTER DATASET

ID	Name	Train	Test
1	Srub	10	751
2	Willow swamp	10	233
3	CP hammock	10	246
4	Slash pine	10	242
5	Oak/Broadleaf	10	151
6	Hardwood	10	219
7	Swamp	10	95
8	Graminoid	10	421
9	Spartina marsh	10	510
10	Cattail marsh	10	394
11	Salt marsh	10	409
12	Mud flats	10	493
13	Water	10	917

of 204 absorption bands. This dataset contains 54 129 samples that can be referred. Table III presents a list of the 16 land-cover classes involved in this study, along with the count of training and testing samples utilized in our experiments.

4) Kennedy Space Center: The Kennedy Space Center dataset was captured by the AVIRIS sensor in Florida, USA. Comprising 614 × 512 pixels, the dataset has a spatial resolution of 18 × 18 m. It includes 176 spectral channels after undergoing a denoising process, encompassing a wavelength spectrum from 0.43 to 0.86 μm. Table IV presents a list of the 13 land-cover classes involved in this study, along with the count of training and testing samples utilized in our experiments.

B. Experimental Settings

We implemented our model on the open-source PyTorch 1.13.0 framework executed on a 64-bit Ubuntu server with 24-GB GeForce 3090 GPU and Intel Xeon E5-2683 CPU with 32-GB memory. The learning rate was set as 0.0001, with the Adam optimizer selected for managing the training process over 8000 iterations. We employed the grid search strategy to fine-tune the hyperparameters on the open-source NNI 2.10.1 framework. Specifically, hyperparameters involve setting β to 0.2 for the graph diffusion and q to 0.5 for contrastive loss function. For the integrating dual view representations, λ_v is set to 0.95. In addition, the coefficients used for L_{PCM} are set to $\lambda_{\text{balance}} = 1.5$, $\lambda_{\text{smoothness}} = 3$, and $\lambda_{\text{recon}} = 8$. The coefficients used for $\boldsymbol{L}$ are set to $\lambda_{\text{uc}} = 0.6$, $\lambda_{\text{sc}} = 1$, and $\lambda_{\text{ce}} = 1.7$.

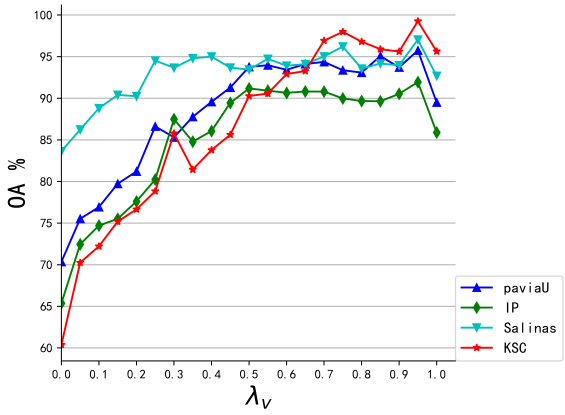

Fig. 5. Relationship between the OA and λ_v on the four datasets.

To evaluate the classification performance of our ACoD-GAT from multiple perspectives, we selected 11 state-of-the-art baseline methods in the field of HSI classification. Specifically, our comparative analysis includes two CNN-based methods (HybridSN [74] and A2S2K [75]), one GAN-based method (ADGAN [76]), one Transformer-based method (SpectralFormer [31]), one manifold learning-based method (AMS-M2ESL [71]), three GCN-based methods (MDGCN [55], miniGCN [56], and SGML [78]), one few-shot learning-based method (DM-MRN [79]), and two contrastive learning-based method (3DAES [80] and S3Net [81]). The comparison models were implemented in their original environments without modifications, and all experiments were conducted in the same environment.

C. Hyperparameter Analysis

1) Analysis for λ_v: To analyze the impacts of λ_v on classification accuracy, we set λ_v from the range of 0–1 with the interval of 0.05 for the four datasets and the evolution of overall accuracy (OA) along with λ_v is shown in Fig. 5. It can be observed that when λ_v is set to 0.95, the best OA is achieved. Therefore, we set λ_v to 0.95 in the following experiments.

2) Analysis for $\lambda_{balance}$, $\lambda_{smoothness}$, and λ_{recon}: λ_{balance}, $\lambda_{\text{smoothness}}$, and λ_{recon} are important parameters in PCM. We set λ_{balance} from the range of 1–5 with the interval of 0.5, $\lambda_{\text{smoothness}}$ from the range of 1–5 with the interval of 0.5, and λ_{recon} from the range of 5–10 with the interval of 1. We analyze the impacts of these parameters on classification accuracy using the University of Pavia dataset as an example, as shown in Fig. 6(a). In addition, we also analyze the impacts of one parameter on classification accuracy when the other two

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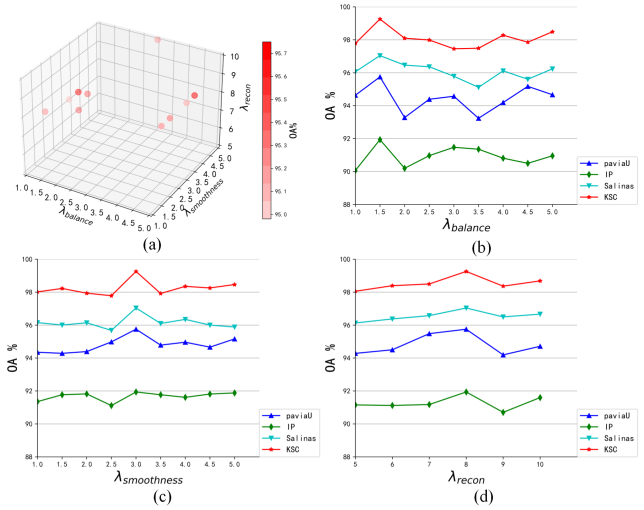

Fig. 6. Relationship between the OA and $\{\lambda_{\text{balance}}, \lambda_{\text{smoothness}}, \text{and } \lambda_{\text{recon}}\}$. (a) 3-D scatter plot of results on University of Pavia dataset. For the sake of visualization, only the top ten results with the highest OA are displayed. (b) λ_{balance}. (c) $\lambda_{\text{smoothness}}$. (d) λ_{recon}.

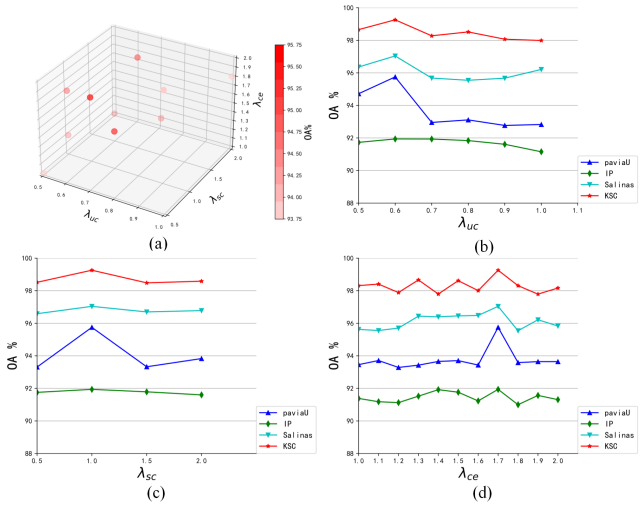

Fig. 7. Relationship between the OA and $\{\lambda_{\text{uc}}, \lambda_{\text{sc}}, \text{and } \lambda_{\text{ce}}\}$. (a) 3-D scatter plot of results on University of Pavia dataset. For the sake of visualization, only the top ten results with the highest OA are displayed. (b) λ_{uc}. (c) λ_{uc}. (d) λ_{ce}.

parameters are fixed, as shown in Fig. 6(b)–(d). When λ_{balance} is set to 1.5, $\lambda_{\text{smoothness}}$ is set to 3, and λ_{recon} is set to 8, the best OA is achieved. Therefore, we set λ_{balance} to 1.5, $\lambda_{\text{smoothness}}$ to 3, and λ_{recon} to 8 in the following experiments.

3) Analysis for λ_{uc}, λ_{sc}, and λ_{ce}: As for λ_{uc}, λ_{sc}, and λ_{ce}, we set λ_{uc} from the range of 0.5–1 with the interval of 0.1, λ_{sc} from the range of 0.5–2 with the interval of 0.5, and λ_{ce} from the range of 1–2 with the interval of 0.1. We analyze the impacts of these parameters on classification accuracy using the University of Pavia dataset as an example, as shown in Fig. 7(a). In addition, we also analyze the impacts of one parameter on classification accuracy when the other two parameters are fixed, as shown in Fig. 7(b)–(d). When λ_{uc} is set to 0.6, λ_{sc} is set to 1 and λ_{ce} is set to 1.7, the best OA is achieved. Therefore, we set λ_{uc} to 0.6, λ_{sc} to 1, and λ_{ce} to 1.7 in the following experiments.

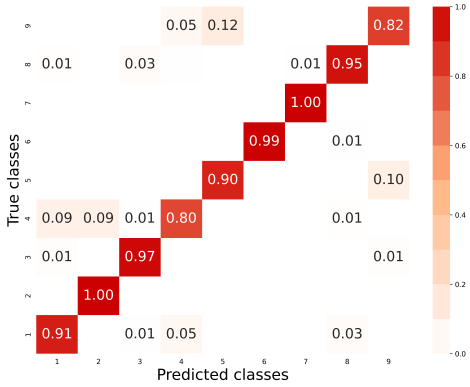

Fig. 8. Confusion matrix obtained by ACoD-GAT on the University of Pavia dataset.

D. Results

Tables V–VIII show the quantitative result obtained by different methods on each dataset, with the highest values in each row highlighted in bold. The corresponding visual results are illustrated in Figs. 8–15.

1) Results on the University of Pavia Dataset: As shown in Table V, the results indicate that ACoD-GAT performs the best in terms of OA, average accuracy (AA), and kappa coefficients. Despite the discontinuous and fragmented nature of the land cover in the PU sample, our method remains effective. The accuracy of CNN-based methods, such as HybridSN and A2S2K2, is comparatively lower due to their inability to extract distance-related spatial information using fixed convolution kernels on regular grids. Due to the strategy of local sampling and training, miniGCN can obtain local optimal solutions and achieve surprising 100% accuracy on multiple classes. However, due to the fusion of CNN and the introduction of global noise, a large number of pixels in Graval and Tree are misclassified into categories that are far apart. Compared with other baseline methods, DM-MRN can achieve higher OA under limited samples due to the simultaneous use of two metrics. The confusion matrix obtained by ACoD-GAT is shown in Fig. 8, it can be seen that there are some misclassifications, such as Trees and Asphalt, but our method performs well in most classes, because it can achieve better performance by utilizing PAM to integrate spatial semantic information at different levels. In addition, compared with other methods, the increases of OA obtained by ACoD-GAT are 14.30% (HybridSN), 15.97% (A2S2K2), 14.32% (ADGAN), 31.56% (SpectralFormer), 10.22% (MDGCN), 9.58% (miniGCN), 20.65% (SGML), 4.98% (DM-MRN), 21.15% (3DAES), 6.39% (S3Net), and 8.40% (AMS-M2ESL).

As shown in Fig. 9, it is apparent that the classification map of SpectralFormer features consecutively large misclassified pixels, which contributes to its poor OA. Similarly, the classification map of 3DAES shows a substantial number of misclassified pixels, underscoring the limitations of solely relying on contrastive learning paradigms, particularly with few labeled examples available. In addition, in scenarios where objects of the same class are scattered across widely dispersed areas, AMS-M2ESL demonstrates poor performance, as its method of cropping HSI disrupts the spatial

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TABLE V

CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE UNIVERSITY OF PAVIA DATASET (%) (TEN LABELED SAMPLES PER CLASS FOR TRAINING)

Class	HybridSN	A2S2K2	ADGAN	SpectralFormer	MDGCN	miniGCN	SGML	DM-MRN	3DAES	S3Net	AMS-M2ESL	ACoD-GAT
1	51.31	75.37	67.72	46.76	69.17	**100.00**	64.13	92.38	84.77	91.96	91.45	91.39
2	89.35	93.93	84.65	60.30	88.56	95.85	69.13	85.43	61.52	92.18	83.93	**99.74**
3	90.90	43.05	64.48	59.65	92.68	1.87	95.55	79.28	63.03	85.93	76.64	**97.27**
4	72.76	62.53	81.37	93.71	91.76	1.83	54.78	98.53	97.19	**99.05**	96.46	79.93
5	99.40	99.77	**100.00**	99.63	96.86	94.01	97.83	99.26	99.03	**100.00**	**100.00**	90.05
6	94.48	80.15	90.60	57.38	80.94	**100.00**	95.14	99.05	95.15	69.10	98.31	99.24
7	95.83	78.66	99.39	85.53	96.60	**100.00**	90.15	99.10	80.00	99.47	99.62	99.70
8	66.67	75.60	73.56	74.13	85.88	**100.00**	88.86	95.60	69.36	81.81	68.74	94.97
9	86.77	84.60	82.28	95.73	96.08	**100.00**	76.95	98.52	82.68	99.79	99.57	82.52
OA	81.45	79.78	81.43	64.19	85.53	86.17	75.10	90.77	74.60	89.36	87.35	**95.75**
AA	83.05	77.07	82.67	74.76	88.73	77.06	81.39	94.13	81.42	91.03	90.52	**92.76**
Kappa	76.09	74.01	76.27	55.60	81.34	81.49	69.20	88.11	68.62	85.96	83.76	**94.35**

TABLE VI

CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON INDIAN PINES DATASET (%) (TEN LABELED SAMPLES PER CLASS FOR TRAINING)

Class	HybridSN	A2S2K2	ADGAN	SpectralFormer	MDGCN	miniGCN	SGML	DM-MRN	3DAES	S3Net	AMS-M2ESL	ACoD-GAT
1	**100.00**	40.45	**100.00**	83.87	97.30	69.44	**100.00**	78.26	**100.00**	**100.00**	**100.00**	**100.00**
2	38.76	69.89	68.19	26.04	74.23	43.30	59.17	71.36	59.87	79.97	**88.36**	86.97
3	52.91	88.87	60.61	21.10	91.24	22.07	81.59	74.94	58.55	72.68	91.22	**95.13**
4	93.40	50.22	98.68	27.93	86.40	33.92	98.24	95.78	95.36	85.90	87.22	**100.00**
5	93.91	95.48	66.17	58.33	93.26	41.01	88.79	86.34	82.40	76.96	**98.94**	83.16
6	96.09	92.26	83.06	51.19	98.06	89.58	96.81	95.75	68.49	99.03	99.86	**100.00**
7	**100.00**	72.00	**100.00**	84.62	83.33	66.67	**100.00**	64.29	**100.00**	**100.00**	**100.00**	87.50
8	**100.00**	98.94	**100.00**	88.12	**100.00**	53.63	**100.00**	97.91	81.38	99.79	**100.00**	**100.00**
9	**100.00**	29.41	**100.00**	80.00	**100.00**	60.00	0.00	50.00	**100.00**	**100.00**	**100.00**	**100.00**
10	59.87	80.24	55.72	71.68	57.51	51.14	54.78	61.63	67.49	79.63	67.67	**86.11**
11	68.36	86.51	69.41	40.66	68.19	64.05	52.43	49.16	51.49	80.74	85.77	**86.67**
12	83.91	66.41	71.18	36.51	86.15	61.92	67.24	59.53	74.03	85.93	**97.60**	92.31
13	**100.00**	71.11	90.77	96.32	**100.00**	84.10	**100.00**	95.12	**100.00**	99.49	**100.00**	**100.00**
14	95.84	97.77	88.29	47.84	97.38	83.51	83.43	95.42	90.83	85.50	98.25	**100.00**
15	92.49	68.02	90.96	47.98	95.50	42.82	99.20	91.19	87.82	97.87	94.68	**99.47**
16	96.23	81.82	**100.00**	97.44	95.29	**100.00**	**100.00**	87.10	**100.00**	**100.00**	**100.00**	98.82
OA	73.20	80.90	74.23	46.10	81.81	58.28	72.05	73.28	69.21	84.17	90.22	**91.94**
AA	85.73	74.34	83.94	59.98	88.99	60.45	80.10	78.36	82.36	90.22	94.35	**94.76**
Kappa	69.77	78.45	71.02	40.05	79.40	52.54	69.30	70.26	65.67	82.01	88.83	**90.84**

TABLE VII

CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE SALINAS DATASET (%) (TEN LABELED SAMPLES PER CLASS FOR TRAINING)

Class	HybridSN	A2S2K2	ADGAN	SpectralFormer	MDGCN	miniGCN	SGML	DM-MRN	3DAES	S3Net	AMS-M2ESL	ACoD-GAT
1	**100.00**	**100.00**	93.75	81.44	99.50	**100.00**	99.85	99.25	**100.00**	100.00	**100.00**	99.75
2	99.95	**100.00**	30.65	96.47	99.68	**100.00**	97.50	99.60	**100.00**	99.95	99.92	99.97
3	**100.00**	99.49	99.85	82.15	97.97	**100.00**	98.32	99.24	98.23	100.00	**100.00**	**100.00**
4	95.95	99.49	98.84	98.77	99.93	**100.00**	99.86	98.92	99.86	100.00	99.78	99.86
5	98.80	99.70	99.89	64.77	90.67	**100.00**	76.76	98.17	83.16	99.33	98.09	94.08
6	98.66	**100.00**	88.48	96.51	97.44	**100.00**	98.84	99.60	99.97	99.95	99.95	98.68
7	**100.00**	99.03	**100.00**	92.88	95.63	**100.00**	99.94	99.55	99.94	100.00	**100.00**	**100.00**
8	77.99	88.81	90.36	49.94	85.41	57.83	67.51	94.40	75.03	80.59	92.44	**98.30**
9	**100.00**	99.18	**100.00**	90.72	99.58	**100.00**	**100.00**	99.76	81.67	99.84	**100.00**	**100.00**
10	87.91	99.38	99.82	73.65	75.22	**100.00**	89.17	98.69	85.88	89.05	96.73	95.14
11	97.64	93.88	**100.00**	90.74	97.93	**100.00**	98.39	97.75	97.00	99.24	98.96	97.46
12	94.57	**100.00**	97.50	95.41	98.07	**100.00**	87.38	99.22	98.34	92.80	**100.00**	99.11
13	**100.00**	99.89	99.45	99.01	96.48	**100.00**	94.26	98.36	98.80	99.78	99.89	94.16
14	99.43	79.79	**99.91**	95.28	81.26	**100.00**	99.25	97.66	96.73	99.06	98.21	98.59
15	72.91	69.45	75.49	85.15	92.41	33.32	85.66	70.40	62.40	**95.30**	77.11	87.38
16	99.78	98.76	**100.00**	67.06	91.12	**100.00**	96.83	99.17	97.34	99.72	99.83	**100.00**
OA	90.51	92.06	92.12	85.00	92.50	82.20	88.46	94.32	85.60	94.29	94.97	**97.04**
AA	95.22	95.43	88.69	79.27	93.64	93.20	93.09	96.86	92.15	97.16	97.56	**97.65**
Kappa	89.45	91.18	87.41	77.12	91.67	84.74	87.28	93.66	84.01	93.67	94.40	**96.70**

topology of the original data. Due to the similarity in spectra between Meadows and Bare oil, almost all baseline methods resulted in misclassification between these two categories, while ACoD-GAT used NAM to divide HSI into regions with similar properties, which is beneficial for enhancing discrimination ability, thus achieving the highest accuracy.

2) Results on the Indian Pines Dataset: Table VI showcases the classification results on the Indian Pines dataset. Due to the serious mixing between different classes of the original Indians Pines dataset, the accuracy obtained by SpectralFormer is relatively low. We observe that CNN-based models (i.e., HybridSN and A2S2K2) achieve higher classification accuracy than SpectralFormer. However, their fixed convolution kernel design restricts the capture of irregular local spatial information. Due to the extensive mixing of numerous nodes from different classes, the performance of GCN-based methods such as MDGCN and SGML is constrained. AMS-M2ESL can better explore complex spectral nonlinear interdependence, performing superiorly in classes with fewer samples, such as Alfalfa, Grass-pasture-mowed, and oats. Despite the use of contrastive learning, 3DAES does not perform well, especially in classes with abundant samples such as Corn-mintill and

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TABLE VIII
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE KENNEDY SPACE CENTER DATASET (%) (TEN LABELED SAMPLES PER CLASS FOR TRAINING)

Class	HybridSN	A2S2K2	ADGAN	SpectralFormer	MDGCN	miniGCN	SGML	DM-MRN	3DAES	S3Net	AMS-M2ESL	ACoD-GAT
1	97.07	98.13	94.41	90.22	46.08	99.07	96.80	96.58	92.25	96.40	96.54	**100.00**
2	39.06	86.52	38.63	60.55	82.01	92.70	87.98	90.53	76.95	78.11	**100.00**	94.14
3	16.26	99.14	36.18	51.08	**100.00**	84.15	**100.00**	95.70	96.48	43.50	97.97	**100.00**
4	19.01	83.58	31.82	7.05	80.49	10.74	99.59	67.06	55.56	87.19	82.64	**100.00**
5	92.72	85.23	89.40	76.47	91.50	16.56	81.46	87.58	64.60	94.70	82.78	**96.08**
6	19.63	83.27	45.21	36.27	90.50	64.84	**100.00**	92.58	92.58	88.13	99.09	92.76
7	52.63	65.79	57.89	57.50	97.94	95.79	**100.00**	90.48	98.10	96.84	98.95	97.94
8	15.68	81.48	26.13	54.19	57.41	55.11	86.22	89.33	50.58	80.52	90.50	**100.00**
9	1.76	99.77	30.00	75.96	94.32	37.45	**100.00**	98.08	91.92	66.47	99.02	**100.00**
10	39.85	95.23	73.10	54.09	68.26	62.18	97.46	86.88	71.53	74.87	**100.00**	**100.00**
11	58.19	92.41	58.92	98.98	72.02	97.31	**100.00**	96.42	96.66	86.31	**100.00**	**100.00**
12	52.74	**100.00**	44.42	63.60	73.64	87.02	79.31	94.83	88.07	87.02	96.35	**100.00**
13	42.86	**100.00**	89.31	**100.00**	**100.00**	97.93	**100.00**	98.92	**100.00**	96.18	99.89	**100.00**
OA	44.52	93.20	60.70	72.68	78.22	75.65	95.06	93.28	85.49	84.41	96.73	**99.26**
AA	42.11	90.04	55.03	63.54	81.09	69.30	94.53	91.15	82.71	82.79	95.67	**98.53**
Kappa	37.71	92.44	55.57	69.26	75.71	72.88	94.51	92.54	83.85	82.64	96.36	**99.17**

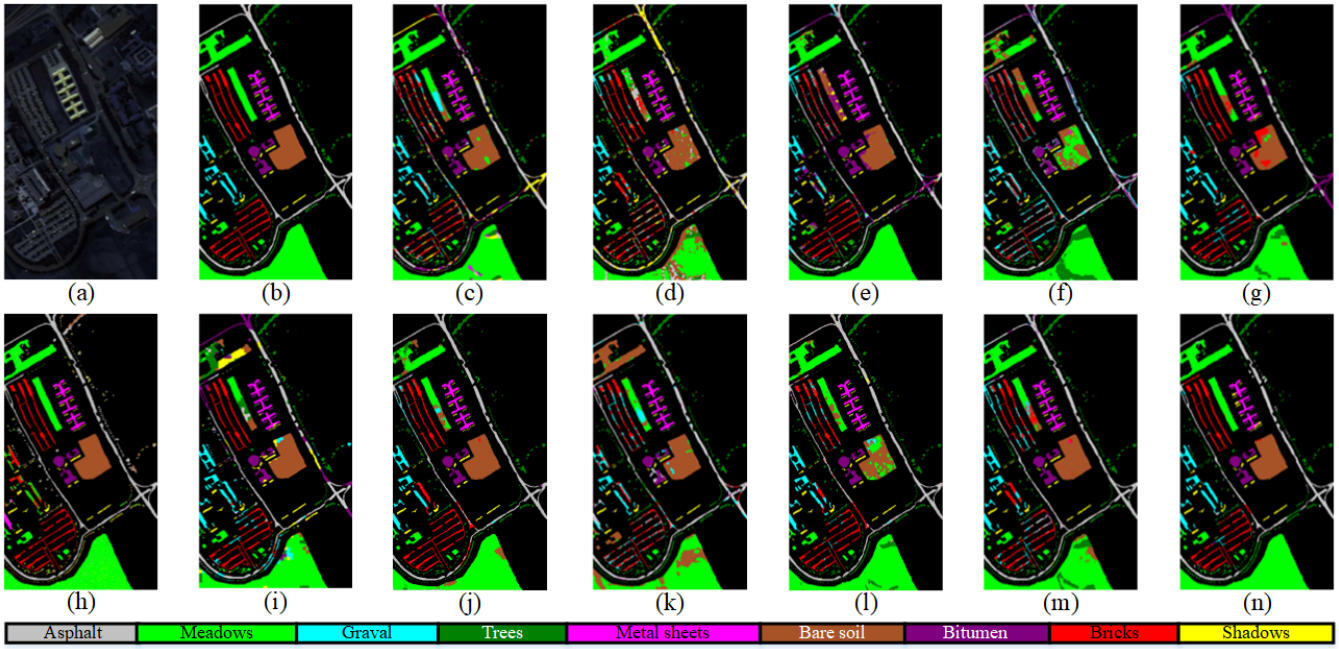

Fig. 9. Classification results of different methods on the University of Pavia dataset. (a) False-color image. (b) Ground-truth map. (c) HybridSN. (d) A2S2K2. (e) ADGAN. (f) SpectralFormer. (g) MDGCN. (h) miniGCN. (i) SGML. (j) DM-MRN. (k) 3DAES. (l) S3Net. (m) AMS-M2ESL. (n) Our method.

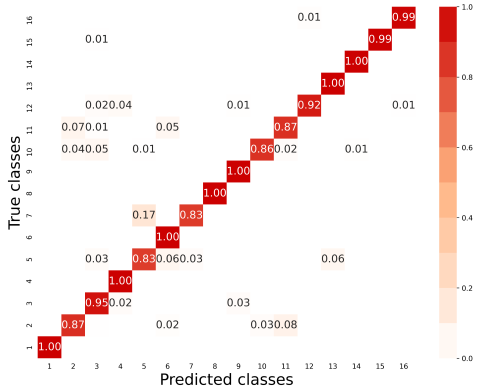

Fig. 10. Confusion matrix obtained by ACoD-GAT on the Indian Pines dataset.

Soybean-mintill. This shortfall arises because 3DAES relies solely on unlabeled examples to pretrain its encoder for feature extraction. S3Net achieved higher OA than 3DAES by utilizing global average pooling and weighted contrastive loss function. Our proposed ACoD-GAT employs supervised and UCL functions for contrastive learning on the dual view, which can improve the model's representation and discrimination ability with fewer training samples. The confusion matrix obtained by ACoD-GAT is shown in Fig. 10, and it can be seen that our method achieves the accuracy of 100% on seven classes. ACoD-GAT exhibits better classification performance in classes with a lower proportion of training samples, such as Soybean-mintill and woods, which demonstrates the reliable capability under conditions of limited labeled samples of ACoD-GAT. In addition, compared with other methods, the increases of OA obtained by 18.54% (HybridSN), 10.84% (A2S2K2), 17.51% (ADGAN), 45.64% (SpectralFormer), 9.93% (MDGCN), 33.46% (miniGCN), 19.69% (SGML), 18.46% (DM-MRN), 22.53% (3DAES), 7.57% (S3Net), and 1.52% (AMS-M2ESL).

Visual classification results are exhibited in Fig. 11. It is evident that numerous pixels contained in Soybean-mintill, Soybean-notill, and Corn-notill are misclassified, largely due to the similar spectral features shared among these land-cover

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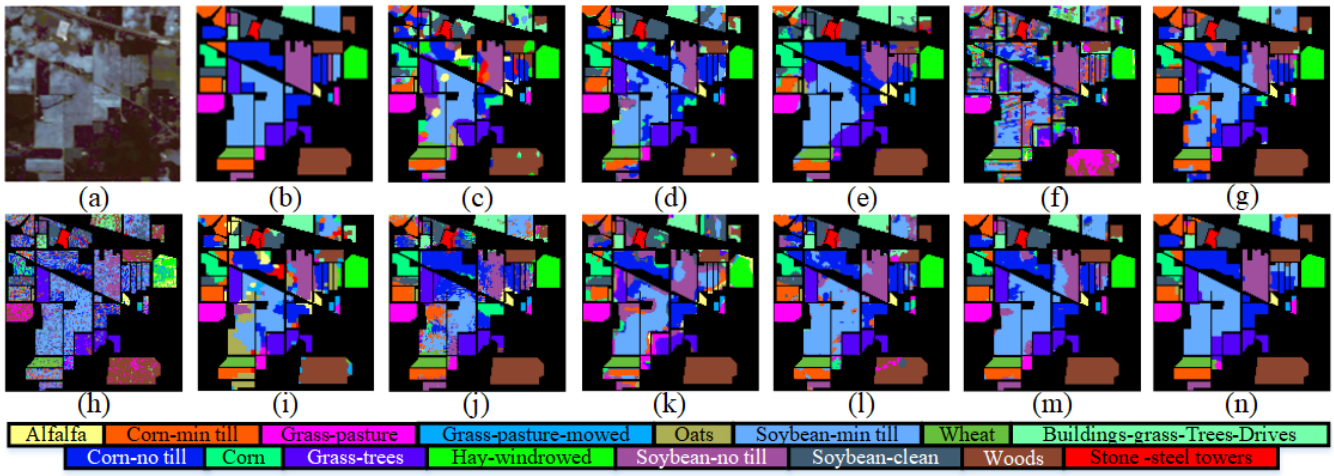

Fig. 11. Classification results of different methods on the Indian Pines dataset. (a) False-color image. (b) Ground-truth map. (c) HybridSN. (d) A2S2K2. (e) ADGAN. (f) SpectralFormer. (g) MDGCN. (h) miniGCN. (i) SGML. (j) DM-MRN. (k) 3DAES. (l) S3Net. (m) AMS-M2ESL. (n) Our method.

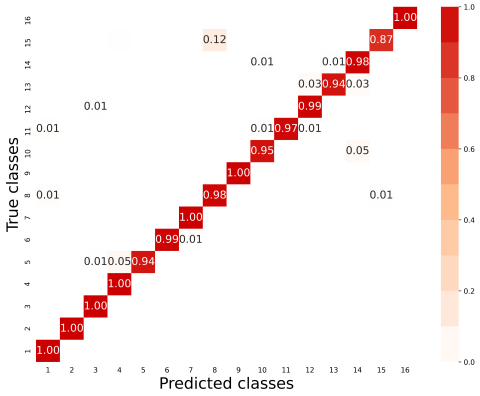

Fig. 12. Confusion matrix obtained by ACoD-GAT on the Salinas dataset.

types. Due to the requirement of inputting a complete graph in the GCN-based method, information from nodes of different classes that are far apart may aggregate during the convolution process, which may lead to misclassification. For example, in the results of SGML, many features that should have belonged to Soybean-mintill were misclassified as Alfalfa. In contrast, our proposed ACoD-GAT exhibits the fewest errors and more smooth classification map.

3) Results on the Salinas Dataset: From the quantitative results in Table VII, it is evident that almost all methods perform better on the Salinas dataset compared to the previous two datasets. This improvement may be attributed to the Salinas dataset having more consistent boundaries than those present in the University of Pavia dataset and the Indian Pines dataset. Similar to the results on the University of Pavia, miniGCN achieved 100% accuracy on multiple classes, but there were many dotted errors on the Grapes_untrained and Vinyard_untrained classes with the highest number of samples. DM-MRN combines relation modules and metric learning to promote intraclass aggregation and interclass separation, achieving good OA. Our proposed ACoD-GAT exhibits the best performance in terms of OA, AA, and kappa coefficients. In addition, compared with other methods, the increases of OA obtained by 6.53% (HybridSN), 4.98% (A2S2K2), 4.92% (ADGAN), 12.04% (SpectralFormer), 4.54% (MDGCN), 14.84% (miniGCN), 8.58% (SGML), 2.72% (DM-MRN), 11.44% (3DAES), 2.75% (S3Net), and 2.07% (AMS-M2ESL). From the visual classification results in Fig. 13, it can be observed that GCN-based methods (e.g., MDGCN and SGML) exhibit numerous misclassifications. For instance, some plots in the "celery" and "stubble" classes are misclassified into classes of plots that are far away from them. This issue arises because these two methods utilize a graph containing all nodes, leading to the aggregation and transmission of information from a lot of nodes of different classes. AMS-MESL employed distance covariance representation and Euclidean subspace projection to enhance spectral–spatial representation capabilities and achieved good OA. However, due to spectral similarity and the lack of considered topological information, many pixels belonging to Grapes_untrained were misclassified as Vinyard_untrained. ACoD-GAT utilizes PAM to integrate multilevel semantic information to accurately perceive different types of HSI regions. The confusion matrix obtained by ACoD-GAT is shown in Fig. 12, and it can be seen that although our method produces some misclassification on Vinyard_untrained, it maintains the highest classification accuracy on Grapes_untrained. In addition, our method can greatly improve representation ability by utilizing supervised and UCL functions and achieved the best OA, AA, and Kappa. The excellent performance on Grapes_untrained and Soil_vinyard_develop with the least proportion of training samples proves the effectiveness of our proposed method in very few samples.

4) Results on the Kennedy Space Center Dataset: Table VIII summarizes the results on the Kennedy Space Center dataset. The Kennedy Space Center dataset, characterized by a high proportion of discontinuous and fragmented land covers, presents significant challenges to the accuracy of HSI classification. The dispersed distribution of regions within the dataset severely limits the contextual information captured by HybridSN during the convolution process, leading to a notable loss of local details and spatial structures, which in turn results in low classification accuracy. Compared with HybridSN, DM-MRN combines convolution modules with relation modules and metric learning, resulting in higher OA.

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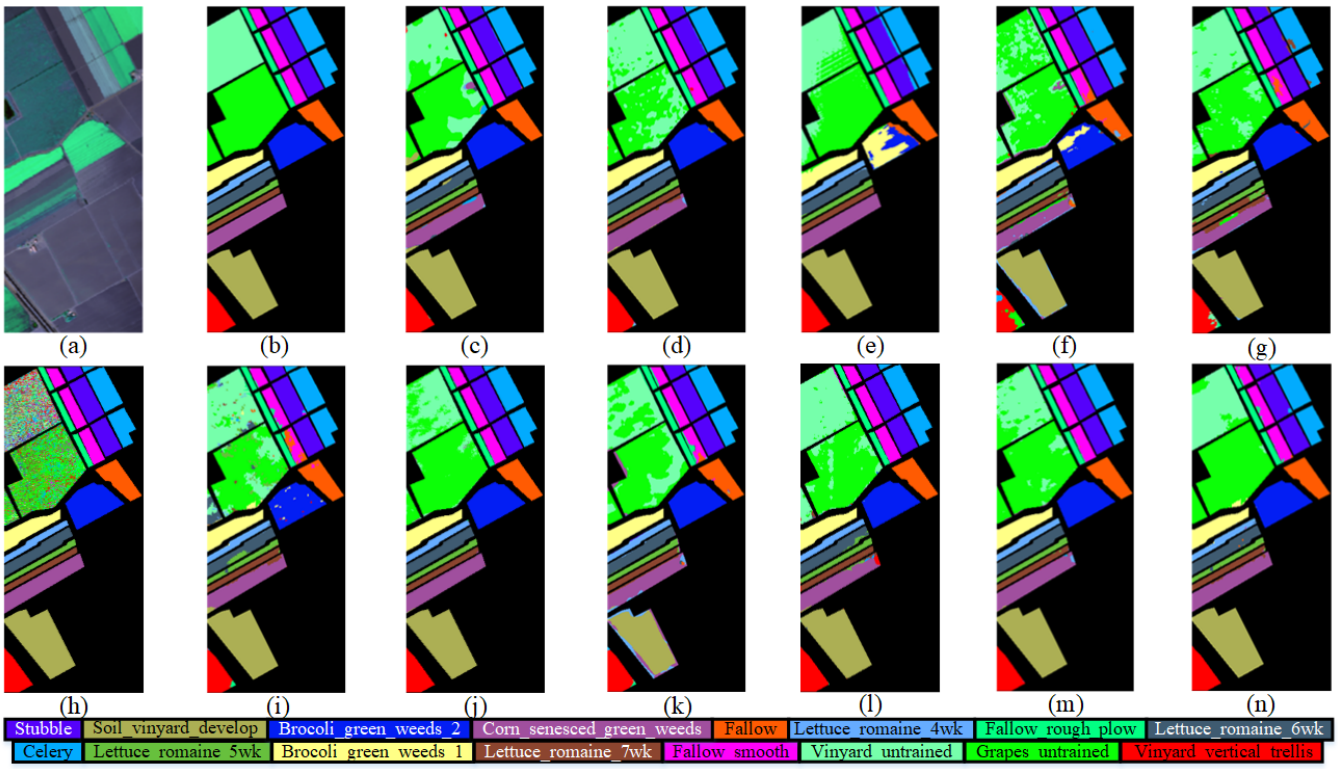

Fig. 13. Classification results of different methods on the Salinas dataset. (a) False-color image. (b) Ground-truth map. (c) HybridSN. (d) A2S2K2. (e) ADGAN. (f) SpectralFormer. (g) MDGCN. (h) miniGCN. (i) SGML. (j) DM-MRN. (k) 3DAES. (l) S3Net. (m) AMS-M2ESL. (n) Our method.

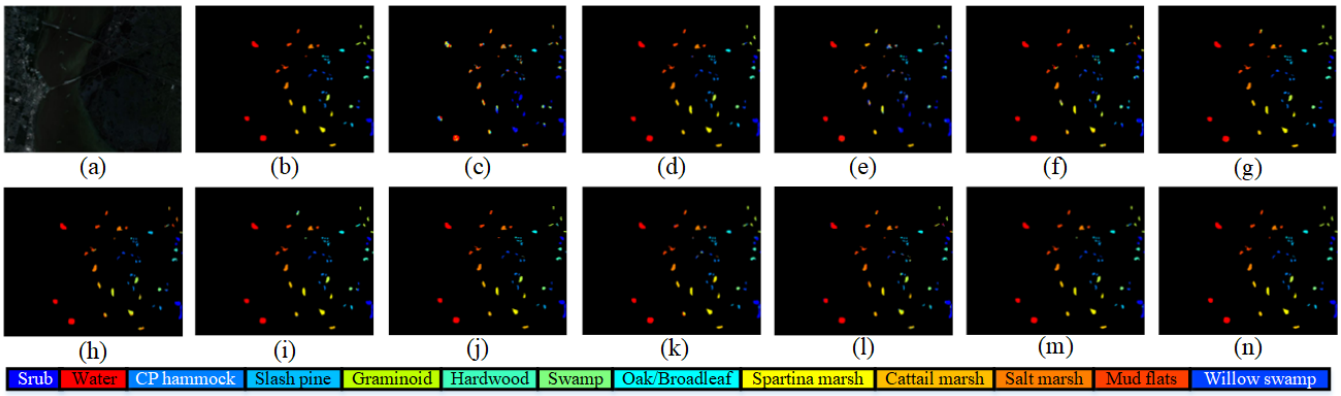

Fig. 14. Classification results of different methods on the Salinas dataset. (a) False-color image. (b) Ground-truth map. (c) HybridSN. (d) A2S2K2. (e) ADGAN. (f) SpectralFormer. (g) MDGCN. (h) miniGCN. (i) SGML. (j) DM-MRN. (k) 3DAES. (l) S3Net. (m) AMS-M2ESL. (n) Our method.

The AMS-M2ESL method attains the highest OA among all baseline methods. Nevertheless, discrepancies in terms of OA remain discernible between AMS-M2ESL and our ACoD-GAT. Our method leverages spatial contextual information from multiple levels and adapts effectively to the complex spatial topologies of various shapes and sizes, thereby enhancing classification accuracy. Employing dual views and conducting training on individual regions, our method substantially curtails the aggregation and transmission of information across nodes of different classes, thereby resulting in obtaining the highest accuracy in ten of the 13 classes examined. The confusion matrix obtained by ACoD-GAT is shown in Fig. 15, it can be observed that our method performs well in each class. The results in Table VIII and visualization results in Fig. 14 illustrate that our proposed ACoD-GAT model achieves higher classification accuracy and superior performance compared to other methods. Notably, ACoD-GAT outperforms competitors in ten land-cover classes, demonstrating its capability to accurately identify discontinuous and fragmented land covers. In addition, compared with other methods, the increases of OA obtained by ACoD-GAT are 54.74% (HybridSN), 6.06% (A2S2K2), 38.56% (ADGAN), 26.58% (SpectralFormer), 21.04% (MDGCN), 23.61% (miniGCN), 4.20% (SGML), 5.98% (DM-MRN), 13.77% (3DAES), 14.85% (S3Net), and 2.53% (AMS-M2ESL).

E. Impact of the Number of Labeled Examples

In this section, the OA of the proposed ACoD-GAT and baseline methods under different numbers of labeled examples is shown in Fig. 16, and we varied the number of labeled examples for each class from 5 to 30 with an interval of 5 and

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TABLE IX
OAs (%) OF THE PROPOSED METHOD BY DIFFERENT SETTINGS ON FOUR DATASETS

datasets	without PAM	without dual view	without UCL	without SCL	without all	ACoD-GAT
PaviaU	91.73	89.50	91.08	90.65	87.34	**95.75**
IP	84.93	85.89	83.50	82.25	81.31	**88.83**
Salinas	94.91	92.68	95.18	93.56	92.12	**97.04**
KSC	95.11	95.64	96.13	95.15	94.97	**99.26**

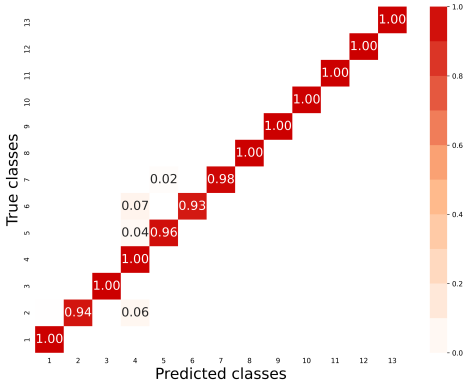

Fig. 15. Confusion matrix obtained by ACoD-GAT on the Kennedy Space Center dataset.

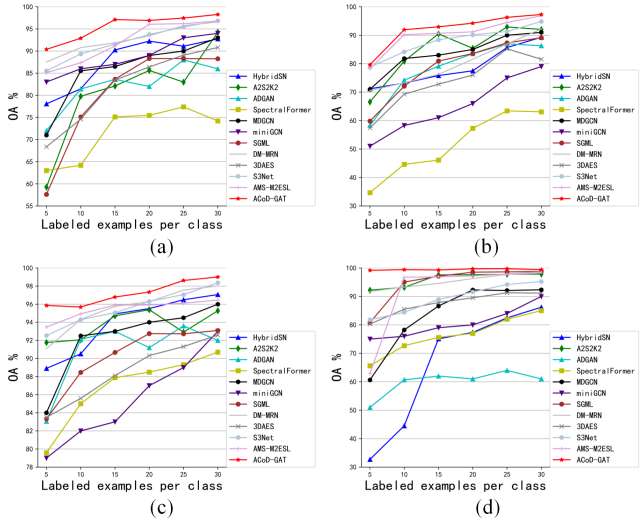

Fig. 16. OAs of various methods under different numbers of labeled examples per class. (a) University of Pavia dataset. (b) Indian Pines dataset. (c) Salinas dataset. (d) Kennedy Space Center dataset.

reported the OAs obtained by all methods on four datasets (i.e., University of Pavia, Indian Pines, Salinas, and Kennedy Space Center datasets). As shown in Fig. 16, the classification performances of almost all methods improve with the increase of training samples. However, due to the training instability associated with GANs [62], the OA of ADGAN may significantly decrease. SpectralFormer exhibits the lowest performance across all four datasets due to the limitations of its deep structure and the adequacy of sample size affecting training and network convergence. As the training samples grow, A2S2K2 shows a rapid improvement in classification performance on all datasets except for the Kennedy Space Center dataset. On the Kennedy Space Center dataset, the OA of ACoD-GAT method hardly increases with the increase of training data. However, in the case of sparse samples, the advantage of our proposed method on OA is more obvious. It can be seen from Fig. 16 that ACoD-GAT achieves the best performance on different sample sizes, which verifies the powerful ability of the proposed framework to fully explore and utilize semantic information at different levels even in extremely limited labeled samples.

F. Ablation Experiments

As mentioned in Section I, our proposed ACoD-GAT incorporates three critical components to enhance classification performance, i.e., PAM, dual view, and contrastive loss function. We demonstrate the effectiveness of these three components by four datasets and report the OAs. Consistent with the experiments in Section IV-C, we employed the same labeled pixels settings. We simplified the model for analysis by designating configurations as "without PAM" and "without dual view" to indicate the removal of the PAM and dual view components, respectively. In addition, considering that the contrastive loss function encompasses both UCL and SCL, configurations are labeled "without UCL" and "without SCL" to represent their respective exclusions. The configuration "without all" denotes the exclusion of all aforementioned components. Table IX presents the results of ablation experiments on four datasets. The results reveal that the removal of different components leads to distinct degrees of impact on the classification results across different datasets. The consistent observation is that the removal of any component invariably leads to a decrease in OA, which indicates that each component contributes to improving the classification performance.

IV. CONCLUSION

In this article, we propose the ACoD-GAT method for HSI classification. In order to fully explore the spatial semantic information at different levels, we designed PAM to establish semantic connections among pixels, objects, and regions, thereby leveraging various levels of HSI contextual information. Building on PAM, we incorporate multiscale manipulation and a diffusion matrix to construct the dual view comprising a local view and a global view and employ GAT to extract semantic information from both local and global perspectives. To address the challenges posed by sparse labeled samples, we design the contrastive loss function that includes the UCL and the SCL, aiming to enhance intraclass compactness and interclass separability by maximizing the mutual information between the same nodes or nodes of the same class across different views. The efficacy of our proposed

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ACoD-GAT is validated through a series of comparative and ablation studies on four real HSI datasets, demonstrating significant improvements in classification performance.

Despite obtaining satisfactory experimental outcomes, we will further improve the proposed method in the following areas. First, we aim to decrease the number of hyperparameters to simplify the training process. Second, the PAM we introduced utilizes various levels of HSI contextual information, and we will explore how to more effectively extract discriminative spectral information. Finally, we will explore how to establish spatial hierarchical relationships in a top–down manner to better utilize the global contextual semantics of HSI.

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Haoyu Jing received the B.S. degree from Zhejiang University, Hangzhou, China, in 2021, where he is currently pursuing the Ph.D. degree with the School of Earth Sciences.

His research interests include remote sensing, deep learning, and computer vision.

Sensen Wu received the Ph.D. degree in cartography and geographic information system from Zhejiang University, Hangzhou, China, in 2018.

He is working as a Professor with the School of Earth Sciences, Zhejiang University. His research interests include spatial–temporal analysis, remote sensing, and deep learning.

Laifu Zhang is currently pursuing the Ph.D. degree in remote sensing and geographical information system with Zhejiang University, Hangzhou, China.

His research interests include nighttime light, urban remote sensing, downscaling, deep learning, and high-resolution data fusion and reconstruction.

Fanen Meng received the B.Eng. degree from Henan Polytechnic University, Jiaozuo, China, in 2020. He is currently pursuing the Ph.D. degree in remote sensing and geographic information system with Zhejiang University, Hangzhou, China.

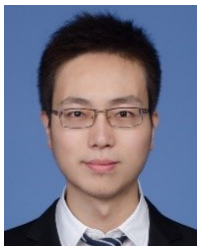
His research interests include deep learning, remote sensing, and image super-resolution.

Tian Feng received the B.Sc. degree from Zhejiang University (ZJU), Hangzhou, China, in 2012, and the Ph.D. degree from Singapore University of Technology and Design (SUTD), Singapore, in 2017.

He was a Post-Doctoral Fellow with the University of New South Wales, Kensington, NSW, Australia, from 2018 to 2019, and a Lecturer with La Trobe University, Melbourne, VIC, Australia, from 2019 to 2021. He is currently a Research Associate Professor with the School of Software Technology, ZJU. His research interests include computer graphics, computer vision, computational design, and human–computer interaction.

Yiming Yan received the Ph.D. degree in remote sensing and geographic information system from Zhejiang University, Hangzhou, China, in 2021.

He is working as a Technician with the School of Earth Sciences, Zhejiang University. His research interests include spatial–temporal analysis and big data mining.



Yuanyuan Wang received the Ph.D. degree in remote sensing and geographic information system from Zhejiang University, Hangzhou, China, in 2016.

He is working as an Associate Professor with Ocean Academy, Zhejiang University. His research interests include spatial and temporal big data management and analysis.

Zhenhong Du (Member, IEEE) received the Ph.D. degree in cartography and geographic information science from Zhejiang University, Hangzhou, China, in 2010.

From 2013 to 2019, he was an Associate Professor with Zhejiang University, where he is a Professor with the School of Earth Sciences. He is also the Vice-Dean of the School of Earth Sciences and the Director of the Institute of Geography and Spatial Information, Zhejiang University. He is also the Director of the Zhejiang Provincial Key Laboratory of Geographic Information System, Zhejiang University. His research interests include remote sensing and geographic information science, spatial–temporal big data and artificial intelligence, and big data and earth systems.

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